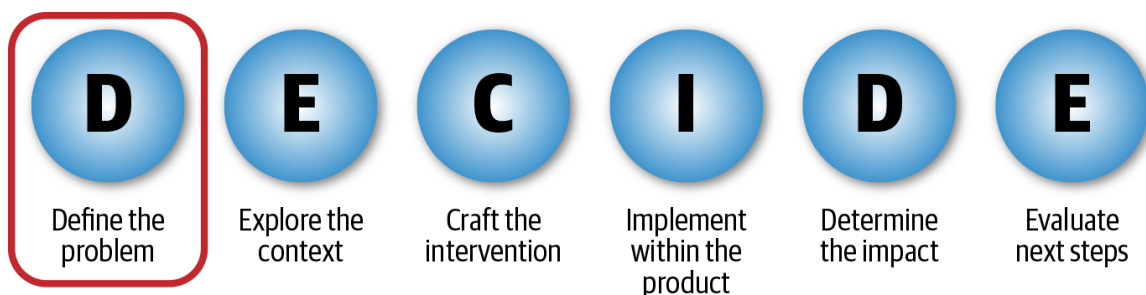


Chapter 6. Defining the Problem



If you currently save for the future, what's your goal? I know it sounds strange, but if I were to ask you in a few hours, the answer might be different. That's because we tend to respond to questions like this with what comes easily to mind, either because it's something we think about frequently or we were recently exposed to.

If I asked you right after you'd gone through an expensive neighborhood, you might answer that you wanted to save up to afford a nice house and lifestyle. If you'd just spoken to your father who was recovering from an illness in the hospital, you might answer that you wanted to have what you needed to pay for medical expenses in the future. But in reality, both are true, and it would be hard to tell which was more important.

At Morningstar, my team thought this might be a problem among everyday Americans—we'd found research documenting it in particular circumstances, such as MBA students not providing stable answers about their goals for a summer internship and major executives not giving stable answers on the goals of their company.¹ If so, it would represent a real challenge to how we encourage people to save and invest, since knowing why people save is key to helping them commit to, and follow through on, their savings goals over time.

We decided to assess the problem among two populations: a nationally representative sample of working Americans over 18, and a subsample specifically of investors. We wanted to determine whether goal instability was in fact a problem worth solving, and if so, test a potential action that would help: using a master list of goals to prompt people to think more broadly.

With an outcome in mind (reducing goal instability), an action (master list), and two target populations, we could design a specific intervention: an online tool to help people analyze their own goals. Indeed, we found that it both seemed to be a major problem and that we could address it. In a randomized control trial, testing different ways of asking people about their goals, we found that when presented with our tool, nearly three-quarters of the participants reevaluated and changed one of the top three goals they'd initially reported, and an amazing 24% changed their top-priority goal.

*Overall, the participants' new goals were longer-term, and more specific, than their initial responses. We published the results and implemented the concept in a software tool for our clients. **This was possible because we started with a clear understanding of the problem, the outcome we sought to drive, and a specific behavioral action that a behavioral actor would undertake.**²*

When Product Teams Don't Have a Clear Problem Definition

I fear that we've all been there: six months into a two-month project, with no end in sight. At the beginning, someone thought everything was obvious and charged ahead. Of course we know what we're doing, let's get on with building it! Six months later, you look back and wonder where all the time has gone and why the product is still off course.

In my experience, the root cause of many bad designs—when designing for behavior change or otherwise—comes from a **lack of clarity from the start**. It comes from a researcher or product manager who had a great idea about

how to fix something and rushed to implement it without thinking through what was really needed. It comes from a product team taking orders from an executive to “just build this” and never really getting the opportunity to learn *why* that was being built or questioning whether it was the right way to do it. And so you end up with a mess—or a misaligned product. Or, as Yogi Berra once put it, “If you don’t know where you are going, you might wind up somewhere else.”³

The problem isn’t unique to designing for behavior change, of course, and there are good tools for problem definition throughout the business world and in the design community. Here, we’re going to take a particularly behavioral approach to this age-old problem. For us, Defining the problem centers on:

The target outcome

What is the product supposed to accomplish? What will be different about the real world when the product is successful? For example, users should have less back and neck pain, with 50% fewer doctor’s visits and physical therapy sessions over the next six months.

The target actor

Who do we envision using the product? Who will do something differently in their lives and thus accomplish the target outcome? For example, sedentary white-collar office workers who don’t exercise regularly.

The target action

How will the actor do it (to the best of our knowledge)? What behavior will the person actually undertake (or stop taking)? For example, users should go to the gym twice a week, for 30 minutes each time.

AN ONGOING EXAMPLE: FLASH

In this chapter, and many of the following ones, we'll show how designing for behavior change works with a running example: an exercise app. Imagine that you work at a B2B company that offers wellness programs to employees of your clients. You already have a number of programs in place, but now your company wants to develop a new product to help people become more physically fit through exercise—especially by going to the gym. You can think of it as Fitbit-style app for a B2B context. The tentative name is *Flash* (to evoke images of rapid, positive change in the user's life).

Your company has seen a suite of wearable computing devices fail in the market and has decided to try something simpler (and less expensive to develop): an app on the employee's mobile phone. In particular, this is the problem definition you're given:

Target outcome

Have less back and neck pain, with 50% fewer doctor's visits and physical therapy sessions over the next six months.

Target actor

Sedentary white-collar workers (employees of client companies), who don't exercise regularly

Target action

Users should go to the gym twice a week for at least 30 minutes at a time.

We'll flesh out this example over time, Exploring the context, Crafting the intervention, etc.—going through the full DECIDE process. As you'll see, this is a *difficult* problem and one that is difficult to build a successful app around. We'll use the example to show the challenges of

designing for behavior change, as well as the process. Along the way, we'll also use other examples to show a range of situations in which these techniques can be applied, and just to keep things more interesting.

Throughout the book, we'll use those three terms, outcome, actor, and action, to Define the problem we're looking to solve. We'll use them first to express the initial *intentions* of the company—the intended target audience, behavior, and outcome. The real world isn't so simple, however. As we learn more about the realities of our users and their situations, we often discover that our intentions aren't realistic. And so, we'll evaluate and refine these three ideas until key stakeholders are on the same page, and potential problems have been identified and resolved. We'll build a clearer and more impactful problem definition over time, centered on these three concepts.

Here's what we'll cover in this chapter, to clearly define the (behavioral) problem:

1. Clarify the overall behavioral vision of the product.
2. Identify who we are trying to serve—the users.
3. Identify the user outcomes sought.
4. Document our (initial) target action.
5. Define success and failure according to that action.

Your company may already have completed some of the initial stages of the problem definition process. If so, you should feel free to jump ahead to the part that's relevant. However, you may find it useful to quickly scan the earlier parts to see if there's anything you missed. At the end of this chapter, I introduce a worksheet that helps with problem definition (“**Worksheet: The Behavioral Project Brief**”). If you can't complete the worksheet, check back for the relevant section on how to handle it.

Start with the Product's Vision

Inspiration for a product or a new feature can come anywhere, from a client request to inspiration that strikes while taking a shower. Either way, you start the problem definition by simply getting the vision of the product down on paper. The vision can be general and somewhat vague—that's fine. It allows us to start the conversation about the more specific, concrete impacts that the product should have—the target outcome.

As an example, let's start with a product vision rooted in the mission of an organization. For example, **the Sunlight Foundation**, a nonprofit that works toward government transparency, gives its mission statement:

Our Mission: The Sunlight Foundation is a national, nonpartisan, nonprofit organization that uses civic technologies, open data, policy analysis and journalism to make our government and politics more accountable and transparent to all.

The vision for each of its products clearly follows from this organizational mission. For example, its Web Integrity Project “monitor(s) changes to government websites, holding our government accountable by revealing shifts in public information and access to Web resources.”

For a company developing health and wellness programs, the product vision might be to “help people take control of their weight and health.”

Nail Down the Target Outcome

After you've recorded your general vision for the product, ask, What should be different about the world when the product is successful? What's the specific, concrete change that should occur because of the product? What could a third party see, hear, or touch? What meaningful thing in the real world *changes* because you've done your job?

The answer to these questions is the product's (or feature's) desired *outcome*. It's the tangible thing that the company *seeks to accomplish* with the product (remember, I use “company” as shorthand for corporation,

nonprofit, or government agency). You might call it a goal, but I like the word *outcome* rather than *goal* because it feels more concrete—it focuses attention on something changing in the world.

Write down that desired outcome (or outcomes).

Clarify the Outcome

Next, let's refine your target outcome with some probing questions. We'll use the example of an environmental cleanup program, in which we move from a vague outcome (decreasing pollution) to one that is clear and more precise. Specifically, we ask:

Which type?

Does the product ultimately seek to change something about the environment (e.g., clean water) or about people?

Where?

What is the geographic scope of the impact (e.g., Chesapeake Bay)?

What?

What is the actual change to the environment or person (e.g., decrease nitrogen pollution)?

When?

At what point should the product have an impact? We're looking for the order of magnitude. Unlike the others, this doesn't need to be precise at this point—"in a few months" or "in five years" is fine.

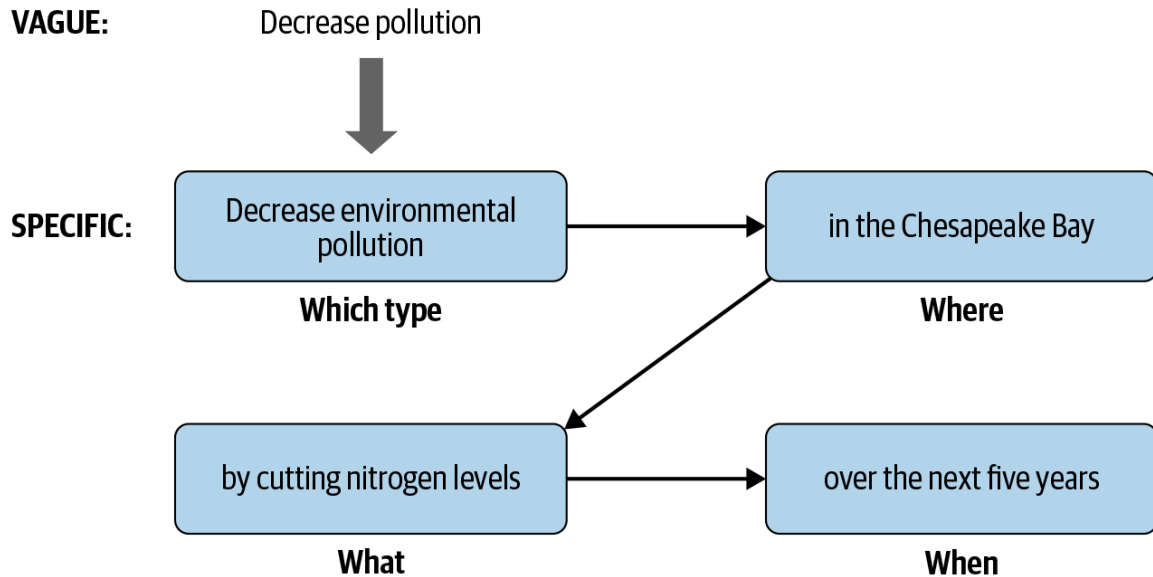


Figure 6-1. How to turn a vague outcome into a specific, measurable one

Write down the answers to these questions with a simple, clear statement that summarizes them. For example, “This product should cause a decrease in nitrogen levels, an environmental pollutant, in the Chesapeake Bay over the next five years.” The new, clearer outcome statement is displayed in **Figure 6-1**.

Based on that statement of the desired outcome, define a metric that you can use to gauge whether the product is successful or not—nitrogen levels in the water or an employee’s weight, for example. You don’t need to settle on the exact measurement yet—but if you can’t define one at all, then the outcome isn’t concrete enough.

Here’s an example of clear outcomes that are readily measurable and ones that aren’t:

Clear, measurable outcomes

Employees with a BMI over 25 will lose 10 pounds. Teenagers in San Jose won’t smoke.

Unclear, difficult to measure outcomes

Users will gain experience with exercise. Users will understand the dangers of smoking.

Having a clear outcome in mind doesn't mean that you're omniscient or that you're locking your company into a strict path for the next few years. Instead, it's a solid point of reference. As problems arise, you can settle disagreements of fact (does this design work or not?) by measuring against the outcome. And, you can settle disagreements of vision (is this the right goal for the product or not?) by redefining the target outcome when needed —*as long as the new outcome is also clearly defined.*

That and nothing else

One of my favorite techniques to refine a team's desired outcome is to ask, "If you got exactly what you asked for (the outcome you've identified thus far), and absolutely nothing else, would you be satisfied?" For example, imagine a team that's working to reduce teen pregnancies and had developed an educational program. They decide that their target outcome is that teens are aware of how pregnancy could disrupt their schooling and future career path. If the teens were educated but didn't care (and did nothing differently), would the team consider that a success? Certainly not. The team would then refine the target outcome to be concrete: they want to decrease teen pregnancy. Education is simply one of the tactics that they *believe* will lead to that outcome.

Vanity metrics—metrics that make a company feel good but don't give an accurate sense of whether the product and company are on the right track—fail on these criteria. For example, consider page views—the stereotypical vanity metric. Let's say a company had *zero* revenue from its flagship consumer product but had lots of page views on the product's website. That usually wouldn't be considered a success. And, vice versa: if the product brought in lots of revenue, but for some reason it had few page views, that *would* be considered a success.

Avoid states of mind

The teen pregnancy example illustrates a common problem that many companies face as they define the product's target outcome: they want to talk about something within their user's heads. Education. Confidence. Skill at doing something. There are two reasons why states of mind are problematic.

First, they are difficult to measure in a consistent and unambiguous way. States of mind can be measured with surveys, but the results of those surveys are highly dependent on how questions are framing, the question order, when and how they are administered, and more. *Highly dependent* means open to debate, misunderstanding, and argument. That's exactly what we're trying to avoid.

Perhaps more importantly, though, states of mind probably *aren't what the company actually wants*. Consider an NGO that trains low-income people in developing countries to be entrepreneurs; do they want the people to know what it takes to be an entrepreneur, or do they actually want the people to start new, successful businesses? If the clients knew how to be entrepreneurs but never actually started businesses, would the NGO consider the program a success? Probably not.

Instead, we want the outcome to be something observable, outside of the individual's head, absolutely unambiguous (to avoid arguments within the company about whether something is successful or not), and easy to measure (so you can gauge success quickly). *The target outcome will define the success (or failure) of the product*. Avoiding states of mind as the outcome doesn't mean that states of mind aren't important; a particular perspective (like wanting to have a bright future without having a baby as a teenager) may be absolutely necessary.⁴ However, it's just not enough.

BEING EFFECTIVE SOMETIMES MEANS BEING CONTROVERSIAL

Often, we focus on states of mind because they are noncontroversial. Everyone can support education; fewer people are comfortable talking about what the education is supposed to do—change people’s behavior, which then changes something in the real world.

Remember, we’re not trying to be noncontroversial. We’re trying to be *effective* at actually changing behavior and helping our users. That takes a measure of self-reflection and honesty—what is your real goal?

Prioritize and combine

If you’re stuck with more than one outcome, that’s fine. Multiple outcomes need extra work, though. First, they need to be organized. If there’s a clear, top-priority outcome, excellent. If not, get the stakeholders together and see if there is a majority opinion on what’s most important. Or, take the list of desired outcomes and ask for each one: would the product still be “successful” if this did not occur? Drop them, one by one.

If winnowing down the list isn’t feasible, there is another, more challenging route. Create an aggregate outcome that combines the contenders for top priority. To do this, you need to get very specific and define a formula that combines them in a way that can get everyone on the same page. This formula is a formal definition of success for the product.

For example, say the two *highest priority* outcomes are “employees will have lower blood pressure” and “employees will lose 10 pounds” A definition of success that combines them both would be “the success of the product is defined as one point for every decrease in average blood pressure across the target population and two points for every decrease in weight, in pounds.”

Unfortunately, most companies just don’t have enough information to understand the complex impacts of the product across related behaviors before the product has been built. Personally, I’d avoid making up a

formula that combines multiple outcomes and settle on one outcome as the top priority.

Avoid stating how the product (might) do it

You may have noticed that there's an important question I didn't suggest asking at this point: how?

For now, try to avoid delving into how the product works its magic (i.e., what action it encourages users to take to make the outcome happen). We'll get to that shortly, after we're armed with additional information.

We focus on outcomes instead of actions for three reasons: first, because there may be many ways to accomplish the action, and the "best" one may not be what comes to mind easily. Second, the link between any behavior and the outcome is uncertain, so we keep our eyes on what really matters: the outcome. And finally, because of "compensatory behaviors." Human behavior is a complex web of checks and balances, not something you just change with a clear effect. A well-known example is moral licensing: sometimes when people exercise, they feel so good about what they've done, they go out and eat unhealthy food. They nullify any benefit that the exercise might have had on their weight.⁵

Why go through the trouble?

Why do we need to define the desired outcomes so carefully and clearly? We want to *pull problems into the present* and resolve them. If the team has a mess of conflicting objectives, or something that can't be measured, then there are problems lurking in the future. The team is likely to argue over what the product should look like as each member tries to meet conflicting, unstated objectives. The head of the company may think the product isn't successful, but the engineers do. The grant funding agency (for NGOs) might cut off funding because it had implicit assumptions that aren't met about what the product would do.

A clear statement of measurable outcomes, which key stakeholders sign on to at the beginning of the project, can resolve many of these problems early.

And, just as with any product development process, finding and resolving problems early on is much cheaper than trying to fix things later.

In addition, a clear outcome statement is essential for future revisions to the product to improve its impact. It forms the basis for measuring the product's success, finding problem areas, and gauging whether proposed changes to the product are worth their salt.

What if no one can agree on the product's intended outcome?

One possible result of this process is that there simply isn't a clearly defined outcome for the product to achieve. Either the stakeholders fundamentally don't agree or the product was poorly conceived and doesn't have real-world outcomes. In that case, the product shouldn't proceed in its current form. In the spirit of failing fast, that's a really good outcome—and should be embraced, painful as it is.

That doesn't mean the team needs to have *consensus* about what the ideal product should do; few companies operate that way. But everyone should know what to expect and sign on to the goal once the decision has been made about the product's intended outcomes. If there are still deep divisions in the team, then problems lie ahead. The team should move on to other interesting products, or change its membership, rather than argue for months over a product that's ultimately doomed.

Define the Metric to Measure Outcomes

Next, you define your outcome *metric*. The metric is what tells you, as unambiguously as possible, whether the target outcome has occurred, and at what level. The outcome metric should flow directly from the target outcome itself. It's how you determine if the outcome is there or not. You should define and write down a formula, even a dirt simple one, which says how the outcome is measured. Here are some easy ones:

Company income

Money received from client companies over the course of a month

User weight

Body weight without shoes, measured in the morning after breakfast

And here's one that is a bit more complex:⁶

Neighborhood connectivity

Number of times users attend social gatherings with their neighbors over the course of a month

Ideally, the metric should be:⁷

Accurate

It actually measures the outcome you want to measure.

Reliable

If you measure the exact same thing more than once, you'll get the exact same result.

Rapid

You can quickly tell what the value of the metric is. Rapidness encourages repeated measurement and makes it easier to see if a *change* in the product was effective.

Responsive

The metric should quickly reflect changes in user behavior. If you have to wait a month until you can measure the impact of a change (even if it only takes a minute to measure it; i.e., even if it is rapid), that's 29 days wasted that you could have been learning and improving the product.

Sensitive

You can tell when small changes in the outcome and behavior have occurred. For the developers among you: floating-point values are great; Booleans are not.

Cheap

Measuring the outcome *multiple times* shouldn't be costly for the organization or it will shy away from measuring the impact of individual changes to the product and have difficulty improving it.⁸

Quite a lot there, right? Yes, but that doesn't mean you need to obsess over the perfect metric. What we're really looking for is a quick check of sufficiency. Treat this like a checklist—for a given outcome metric, ask: is it specific enough that there won't be too much disagreement when it's measured? Is it reliable enough that it won't fool the team into thinking the product is working when it actually isn't?

Working with Company-Centric Goals

Thus far, we've talked about a product development process that's focused primarily on the user and what the product can do for them. But I've found that companies sometimes take two very different approaches to behavior change. They can either:

- Focus on how the product will *benefit the user*, which in turn helps the company's bottom line
- Focus on how the product will *benefit the company*, by way of providing value to the user

This difference is in how companies think about the *value* of changing behavior; it isn't related to the behavior itself. The target behavior could be inside the product or outside the product, socially important or trivial; that doesn't matter as much.⁹

In the first case, which we'll call a *user-centric approach*, the company might have a vision of improving financial wellness (like Acorns or Mint) and need to figure out what actions users can reasonably take that will best make that happen. In the second case, a *company-centric approach*, a company might have a purely self-interested goal, like increasing customer renewals, but needs to provide real value to the user in order for that to succeed. The second approach includes red-blooded capitalists as well as NGOs that need to make their case to their funders. There's nothing wrong with that approach—as long as the companies build products that people like. But it does mean that the process is different.

In the user-centric approach, the process of Defining the behavioral problem is this:

Product Vision (for the User) → User Outcome → Actor → Action

In the company-centric approach, we add another step:

Product Vision (for the Company) → Company Objectives → User Outcome → Actor → Action

The product vision is *why the product is being developed*, at a high level; and the company objectives are *what the company seeks to achieve*, for itself, by building that product.

Since the previous section highlighted examples of the user-centric approach, let's now walk through the company-centric process (you can safely skip this if you're using a user-centric approach).

State the vision

As before, the Definition process starts by writing down the high-level vision the company has for the product. That vision, however, should answer how the product will generally benefit the company. For example, this product should:

- Expand the company's appeal into new markets.
- Increase revenue.

- Demonstrate the expertise and capacity of the organization to take on new projects, to support new grant funding.
- Increase public awareness and interest (or brand prestige) of the organization.

State the company's objectives

With the company's vision for the product in mind, translate that vision into one or more specific, measurable objectives that benefit the company. For example, ask:

- How will the product be judged a success or failure at fulfilling the company's vision? How will success be measured?
- What would a third party observe about the company that's different because of the product? Increased retention of customers? Increased upsell? More referrals of new customers?

The company's objective might be to win 35% of the wellness program market among technology companies in the next year. Or it might be to win at least one million dollars of additional grant funding for the next year. Write down that initial company objective.

Based on that initial statement of the company's objectives, fine-tune it by asking who, what, when, and where the outcome occurs (as we did in the previous section).

Define the user outcomes

Building your business or establishing your expertise to funders is great, but your users probably don't care. Sorry. You need to deliver something of value to them. Without that value, you can't meet your business objectives.

So, putting aside the financial (or other self-interested) goals of the product for a moment, what does the product mean for users? We want to define the measurable changes in the world that are caused by the product *that the user would care about*.

Here are some questions to help draw out those outcomes:

- What does the product deliver? What's its core value proposition as the user would see it and measure it?
- After users have used the product, what's different about the world?
- What tangible thing would a user look at (or hear or see), sometime after using the product, and say, "Hey, I want to use that product again"? (The product itself doesn't count, sorry.)
- How would you know that your users have gained the maximum value from your product?
- What do the users *do* because of the company's increased brand awareness?

For example, consider the company objective of winning 35% of the wellness program market among technology companies in the next year. The specific user outcome might be what we identified above for our *Flash* app: the product helps the user decrease doctor and physical therapy visits by 50% over six months. Or the product might help the user drop two waist sizes.

In the previous section, we discussed a set of rules and tips for clarifying target outcomes. All of them apply for company-centric goals as well: avoid states of mind, make sure the outcome is measurable, find disagreements early, and don't obsess about *how* the product will achieve these outcomes yet.

A Quick Checklist

In summary, this is what makes a good outcome, both for the user-centric and for company-centric goals:

- The outcome is *what will be different* in the real world when the product is successful.

- The outcome should be something *tangible*, and not something in the user's head. Often the user's head is just a proxy for what the company really cares about—a tangible outcome that occurs because the user's knowledge or emotions have changed. For example, lower BMI, or pounds lost, instead of knowledge about the importance of exercise and maintaining one's weight.
- The outcome should be something *unambiguously measurable*. For example, say your product is supposed to “decrease government corruption.” What is corruption, exactly, and how is it measured?
- The outcome should be able to *signal success*. If you can say, “Well, if X didn't happen, the product would still be a success,” then X is not the outcome we're looking for.
- The outcome should be able to *signal failure*. You must be able to think of a plausible case in which the outcome would indicate that the product is failing.

Who Takes Action?

Usually the company's business goals and market research specify who a product is supposed to serve: the people who buy the product (private company) or the people who they are tasked with helping (NGOs and government agencies). Here, we need to dig a bit deeper. We want to know not just who the product will serve, but who is it that will take action? Whose behavior are we seeking to change?

Often the person being served and the person taking action are the same: user = actor. But that's not always the case. The user may influence another person, who then does the real work. This is common in situations like B2B sales (where the user is a different person than the buyer) and with advocacy websites that try to influence policymakers to change regulations (like on the cost of gasoline), which then change behavior in society to drive outcomes (like lower greenhouse gas emissions).¹⁰ For the sake of

simplicity, though, and to use language people are familiar with, I'll assume they are the same for this discussion.

Write down the target actor as specifically as possible: age, gender, location, number of people, and so on. It can also help to write down who is *not* being targeted; for example, people without smartphones, the wealthy, or expats.

In all likelihood, some target actors aren't a good fit for the product, and you'd waste your time trying to target them. True. For now, we want to know the *potential* actors you should investigate further. You may end up targeting only a subset of them.

If the company has no idea who it is trying to serve and who needs to take action, then it's back to the basics. A behavioral product, like any product, must serve a user need. It can only help people change their behavior to the extent that they care about the product at all. To get a handle on the unmet user need, you need a traditional market research or (non-behavioral) product discovery process. That's beyond the scope of this book; from here on, I'll assume you have a sense of the target user already.

Document Your Initial Idea of the Action

Thus far, we've intentionally not talked about how your actors would accomplish the outcome. That's to help us separate and clearly think about the outcome, without assuming a particular way to accomplish it. But, of course, we have ideas about what action the person should take. We need to record that idea. It won't necessarily be correct—but by writing it out, we'll better be able to evaluate and refine it.

What defines an action? I think about it as two blanks you want to fill out in this sentence:

Our product will help the user [start/stop] doing [describe action].

For most companies, especially those with existing products on the shelf, the universe of possible user actions is tightly constrained by the company's

business model, product strategy, and internal culture. At HelloWallet, for example, we looked for actions that would be appropriate for a wide range of users and weren't being well covered by existing products, that fit our company mission. So job-hunting tools were out, as were mortgage finders and pure lead-gen tools.

Next, ask specifically, "How does the action cause the target outcome?" The action should directly and clearly cause the target outcome. If the current action doesn't do that, is there another, later action that *does* cause (or is a missing piece that's required to cause) the outcome? If so, focus on that one instead. We want to target a behavior that directly supports the outcome.

For example, let's assume we want to drive community volunteerism. The action "Go to a seminar about the importance of community engagement" *might* lead people to get more involved in their communities. The direct link between the action (go to seminar) and outcome (community members volunteer) is a bit tenuous though. What if the person doesn't pay attention? What if they've been forced by their spouse to attend such events in the past? A better, more immediate action is "Volunteer at a local soup kitchen." That's the real point of the seminar and has a direct link to the outcome we care about. A seminar might be a useful tactic along the way, but it's only a tactic to support action—it's not the target action itself.

Clarify the Action

As we did for the company's target outcome, we're looking for a concrete, specific definition. A physical, measurable action someone will take. Avoid actions that just affect the person's mental states (reading educational material) and dig deeper into what the person does with their new education that causes them to *do* something differently and achieve the outcome.

For example:

Outcome sought

People don't get lung disease.

Vague action

Users avoid cigarettes. (Does that mean they cut down on smoking? Go cold turkey?)

Action that's too far removed from the outcome

Users attend a seminar about the dangers of smoking (OK, but do we care if they attend? Or that they actually stop smoking?).

Clear action

Users don't buy cigarettes at all.

You'll notice that the action doesn't specify how exactly the product will help the user not buy cigarettes. It could be by helping them avoid stores where cigarettes are sold, or by decreasing the desire to smoke with nicotine patches.

Like the target outcome, the target action isn't written in stone. It should be clearly defined so that it can be built into the product and clearly measured. The definition, and measurement, will help with fine-tuning the product. They'll also help with revisiting and revising the target action itself, if the need arises.

A Metric for Action

As we did with the outcome, we want to translate the target action into a specific metric that will assess whether people are taking action. The action metric tells you whether (and to what degree) the user is taking the target action, the action that is supposed to drive the desired outcome. If the desired outcome is a specific level of weight loss and the action is exercise, a sample metric would be "How much is the user exercising, and how often?" A good action metric must pass the same tests as the outcome metric: accurate, reliable, rapid, responsive, etc.

The action metric covers *what* is measured, *how* it is measured, and for *how long*. For example, one way to define purchase behavior for a particular product is: money spent (not flexible future commitments) through product sales and subscriptions over a 30-day period.

It also needs to be specific, because if the value changes over time, specificity helps us determine if it changed because of your product. If the definition is unclear, a change in the data may be caused by a change in how people interpret or measure it.

Here are two examples of action metrics:

Action: “the user exercises”

Bad metric

User exercises = how much the user reports exercising each day. This is a bad metric because (a) the user may not know how hard they are exercising without the help of a timer, heart rate monitor, or other tracker and (b) the user may stretch the truth.

Good metric

User exercises = how long and how hard a heart rate monitor automatically tracks the user exercising each day.

Action: “the user studies a new language”

Bad metric

User studies = an expert evaluation of language proficiency on a lengthy written exam. This metric is problematic because it focuses on the intended *outcome* rather than on the *activity* that

we assume, rightly or wrongly, leads to that outcome. It also takes a long time to measure and can't be measured frequently (without annoying the users!).

Good metric

User studies = time spent within the application, or number of exercises completed with a minimal level of accuracy.

Clearly there are trade-offs when creating action metrics. The most accurate metric may take too long to gather, or the cheapest metric may not be reliable. Again, there's no need to obsess over it—we're looking for an action metric that is sensitive enough to quickly show when there's a problem and accurate enough to not mislead the team.

Look for the Minimum Viable Action

The *minimum viable action* (MVA) is the shortest, simplest version of the target action that users absolutely must take to so that you can test whether your product idea (and its assumed impact on behavior) works.¹¹ It builds on the Lean start-up concept of the minimum viable product: **the smallest set of features that allows the product to be deployed and tested in the field.**

How do you find the MVA? Look over the action you've proposed. I think of the minimum viable action as something that you arrive at by cutting back on what naturally came to mind the first time. Cut back from the obvious until only the necessary remains:

1. *Cut repeated actions down to the first action.* If you have a repeated action, can you start by supporting a one-time action? One-time actions are easier for users to perform and engineering teams to build than repeated ones, all else being equal. And they still provide valuable insight into whether the software works at supporting the behavior. For example, if you want to help people

lose weight by getting them to run two miles twice a week, see if they will run or jog *at all* before trying to change their regular routine.

2. *Cut big actions down to simpler ones.* Even if the target outcome won't be achieved initially, can you cut the action down into something smaller and shorter but still fundamentally the same basic task? The goal is to test the core premise, as with a minimum viable product (i.e., instead of asking people to run frequently, can they start with a group jog?)
3. *Drop steps in the sequence altogether.* Can you identify the high-risk, most uncertain aspects of the action (i.e., having people feel comfortable running alone in a new place) and either remove them altogether from the target action (e.g., run at work with colleagues) or field test them before developing the lower-risk aspects? Similarly, are there steps that are just nice to have and can be removed?

I find that people don't naturally think about the smallest possible action to change behavior. We like to think big. That's fine. It's useful, good, and most natural (i.e., easiest) to express that big vision first. That provides a blueprint that the team can go back to and draw upon as the product develops.

However, once those big behavior change thoughts are up on a board, and we see all of the pain we're thinking of putting our downtrodden users through, then reality should hit—the more work that users have to do, the less they are likely to do it (with some important exceptions I'll cover later). Hence, the MVA.

To demonstrate the idea, let's use an example of helping users learn Spanish. Here are actions that the team might target:

- Complete an online training course.
- Visit Spain for a few weeks to be immersed in the language.

- Label each item in the household with their Spanish names.

With simpler MVAs, you can test the core assumptions of the approach and its impact more quickly:

- Complete a single module of an online training program.
- Get a Spanish-language conversation partner who is committed to only speaking Spanish with the user.
- Label a few items in daily use with their Spanish names.

OK, we have a rough idea of the potential action that users might take. The next chapter will dig deeper into exactly who the users are and whether that action is realistic for them to take.

A Hypothesis for Behavior Change

You now have all the things you need to determine what success and failure would mean for your product *before you build it*. You know who the product is supposed to serve. You know what real-world outcome that action is supposed to cause. You have an initial sense of the action you want to drive in order to accomplish that outcome. You don't have all of the details, of course, but that's OK. At this stage, you have the rough outlines you need.

Write down a sentence that says what the product is supposed to be doing, and for whom. For example:

By helping overly sedentary white-collar workers (actor) to go to the gym (action), our product will cause them to have less back and neck pain and go to the doctor and physical therapist 50% less often (outcome).

As other authors have thoughtfully noted, we can think about this as our *hypothesis for behavior change*.¹² The general format is:

By helping the [actor] [start/stop] doing [describe action], we will accomplish [outcome].

It's a nice way to structure the outcome-action-actor, and it reminds us that everything we plan to do is just a plan. It's full of assumptions about the real world, and calling it a hypothesis (which it is) helps us remember that it's something we need to test to see if, in fact, we're right and we'll have the effect we hope.

Now, from your user research about what's feasible for users to accomplish and from your market research about what will differentiate and sell the product, you should be able to add in specifics about the proposed product and its impact on your business. For example:

By helping 25 to 35-year-old white-collar employees of tech companies in urban areas (actor) to go to the gym twice a week for at least 30 minutes (action) the product will cause them to decrease their doctor and physical therapy visits for back or neck pain to 50% less than they otherwise would have had over the following six months (outcome). When successful, it should double our current revenue (company objective).

In this statement, the team is saying: if this happens, the product will be a success; if not, it will be a (complete or partial) failure. Later on, we'll translate this statement into a set of metrics to gauge whether the product actually succeeded or failed at its goals.

Our goal isn't to create a false sense of security by thinking we can forecast the future and how the product will actually play out. There are lots of assumptions built into this definition of outcome, actor, and action. We want to draw out those assumptions, to get something that can be explicitly tested and then revised as lessons are learned. Once they're on paper, we can review and improve them—through pre-mortems, thinking-hat reviews, and such.¹³ And, perhaps most importantly, it helps us fail fast—to make sure that the key stakeholders in the company are on the same page before building the product. If they aren't, now's the time to fix it.

Defining success and failure ahead of time doesn't mean that we can't change the goalposts now and then. As we learn more about the market, product, and the company's other opportunities, our understanding of

what’s “good enough” will change. But make sure the team knows about the change and understands the rationale. No one likes a moving target—especially when that means making the job harder for the team midstream and without explanation (raising the standard) or it seems like accepting failure (lowering the standard).

Examples from Various Domains

Desired outcomes and target actions can be a bit abstract, especially given the tremendous range of possible products that can affect user behavior. So let’s look at some concrete examples (Tables 6-1 and 6-2). Since the approach is somewhat different for user-centric versus company-centric products, I’ve broken them up into two different tables. I’ve started both tables off with our sample product—the Flash running app—to show how the analysis would proceed from each perspective.

Table 6-1. User-centric examples

	Example 1	Example 2	Example 3
Product	Flash, an exercise app	Financial wellness app (Acorns)	Cigarette Cessation
Vision	Help people take control of their health	Provide broad access to financial guidance	Help smokers quit and avoid cancer
Outcome	Less back and neck pain (fewer doctor and physical therapy visits)	Americans have sufficient emergency savings	Cigarette smokers stop smoking
Actor	White-collar technology company employees	Gig economy workers in San Francisco	Long-term smokers who want to quit but have failed with nicotine patches
Action	Go to the gym twice a week for at least 30 minutes	Users automatically transfer money to savings account each month	Smokers switch from cigarettes to vaporizers and decrease total nicotine intake by 50%

Table 6-2. Company-centric examples

	Example 1	Example 2	Example 3
Product	Flash, an exercise app	Grocery store website	Breathalyzer
Vision	Expand into adjacent wellness markets	Expansion of business into upscale grocery shoppers	Make company's breathalyzers standard in cars across the country
Company objectives	Increase revenue from corporate wellness program clients	Double number of upscale buyers	Attain 25% market share in three newly entered states
User outcome/user need	Less back and neck pain (fewer doctor and physical therapy visits)	Learn how to cook healthy meals	Prevent accidents and tickets from drunk driving
Actor	White-collar technology company employees	Upper-income Denverites commuting from the suburbs	Drivers in the targeted states, who have had a suspended license, a ticket, and/or an accident from a DUI in the past
Action	Go to the gym twice a week, for at least 30 minutes	Users take free cooking classes on grocery website	Stop default driving home behavior 75% of the time after heavy drinking, using company's breathalyzer before starting the car

Reminder: Action != Outcome

Despite our best efforts, there may be a big leap between the outcome and the action. Will transferring small amounts of money into a savings account (when the user can choose to transfer it back or spend it) really solve the broad lack of emergency savings in the United States? That's a tough question to ask. The purpose of this chapter was to provide a clear direction for the product development process, uncover hidden assumptions, and, sometimes, determine that a pivot is needed, and a different approach or behavior should be targeted.

Sometimes the outcome that's important is, in fact, an action that the user takes. The first example could easily have had “users exercise” as the outcome, as well as the action. This occurs when the action itself has an unambiguous, real-world outcome (like exercise). However, I argue that companies should avoid equating the two in most cases. It's an easy way to hide assumptions about why the action is important and thus to potentially choose the wrong action.

Putting It into Practice

This chapter started off the DECIDE process by Defining the problem; in particular, with the outcome we're seeking to achieve, for whom, and our initial idea of how (an action). Here we summarize the key lessons and show you how to use the exercise in the workbook.

Here's what you need to do:

- Define the real-world outcome that the product should have. Avoid states of mind—focus on measurable outcomes that define the success or failure of the product.
- Translate company-specific goals (e.g., increased profit) into real-world outcomes that the user actually cares about.

How you'll know there's trouble:

- The company can't agree on the intended outcome of the product.
- The company knows what it wants but doesn't offer something that users care about.

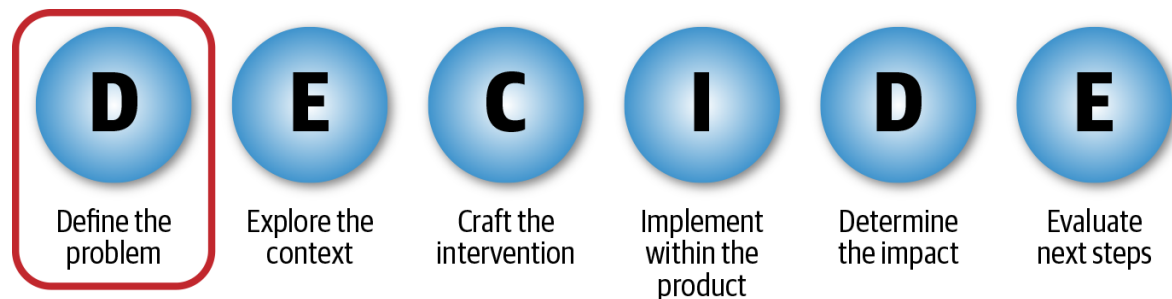
Deliverables:

- The behavioral project brief: a clearly defined outcome, a clearly defined target population (actor), and an initial idea for the action the actor will take. You can bring all of these together into a *hypothesis for behavior change*.

Learning about a technique often isn't enough on its own. Perhaps the biggest obstacle readers faced with the first edition of *Designing for Behavior Change* was that they were unclear on how exactly to act on it, and that kept them from getting the most out of the book.

For Defining the problem and each of the other steps in DECIDE, we have exercises your team can go through to implement the process. For all of the exercises, we'll use a consistent example throughout: the Flash app introduced in the sidebar at the beginning of this chapter. In this example, your company provides wellness software to employers, who then provide it to their employees. You're working on a mobile wellness app to help people exercise—it'll be a new product for your company, and you're just starting on the project.

Worksheet: The Behavioral Project Brief



Goal: Understand and articulate the aims of your team's new product—the outcome, action, and actor it should target.

Project: Flash app

- ☒ New product, feature, or communication?
- ☐ Change to existing product, feature, or communication?

Vision: Briefly describe *why* you want to change behavior and how this product fits in.
Flash will help employees take control of their health.

Outcome: What do you hope to achieve with the product? Consider both the company's objective, as well as the real world, measurable change users will see and value. Then, drill down and define a rough metric your team can use to evaluate the product and an idea of what success looks like in numbers.

Company objective: Increase revenue from corporate wellness program clients.

Real-world outcome: Less pain (back, neck, etc.).

Performance metric: Doctor and physical therapy visits

Definition of success: 50% decrease in doctor/PT visits

Actor: Who is the specific user (or person involved in the product) who causes the outcome?

Who is the actor? Sedentary white-collar workers

Action: What does the actor do/stop doing to accomplish the outcome? This an initial idea; we'll refine it later.

What is the action? Go to the gym twice per week

A Hypothesis for Behavior Change: Alternatively, you can write out this information as an explicit hypothesis to remind the team that nothing is for certain, and you'll need to test that hypothesis in practice through the product:

By helping [actor] sedentary white-collar workers ☒ Start ☐ Stop doing [action] go to the gym twice per week, starting with 30 minutes per session, we will accomplish [outcome] decreased pain and a 50% decrease in doctor or PT visits over six months.

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- 1 Bond, Carlson, and Keeney (2008); Bond, Carlson, and Keeney (2010)
 - 2 Detailed results from this study can be found in Sin, Murphy, and Lamas (2018).
 - 3 Yogi Berra was a professional baseball player and coach, known for his (mal)aphorisms. For example, see *"Yogi Berra Dies at 90"* in the *LA Times*.
 - 4 My thanks to Brian Merlob for pointing this out.
 - 5 Again, h/t to Brian Merlob for this one.
 - 6 Originally inspired by the (now-closed) community connections start-up, Neighborsations.
 - 7 There are a variety of perspectives on what makes a good metric but no generally accepted and applied definition. These are characteristics that I've found to be important.
 - 8 It may take an up-front investment (that isn't cheap) to make recurring measurement cheap. We want to set up data collection that will be cheap and easy to check whenever there is a change to the application. Survey data, for example, is often "cheap" to measure the first time, but the cost usually remains the same with each iteration (and survey data is plagued with biases, as we discuss in *"Figure Out How to Measure the Outcome and Action by Hook or by Crook (Not by Survey)"*). Ideally, we want automatically gathered administrative data—collected from the original source without human intervention or extra costs. Asking people

what they spend money on is a survey. Their actual credit card transactions are administrative data.

- 9 In other words, this is a different distinction than the one I made in the [Preface](#) about behavior change affecting a behavior *within a product* or *outside of it*. In either case, a company could start with the benefit for the user or the benefit for the company.
- 10 My thanks to the folks at ForumOne for pointing this out.
- 11 In the context of habits, BJ Fogg has a similar idea in his new book, *Tiny Habits* (Fogg 2020).
- 12 My thanks to Rajesh Nerlikar for first introducing me to the idea. Matt Wallaert has a similar idea in his book, *Start at the End* (Wallaert, 2019).
- 13 On pre-mortems, see Klein (2007)—h/t Paul Adams. On thinking hats, see De Bono (2006). There are also many assumption-mapping techniques to draw out (and question) assumptions in a decision or process. As discussed in [Booth \(2019\)](#), Shopify has a fun one called the *assumption-slam*—h/t Anne-Marie Léger.